

```
ansible/inventory/kvm/
/playbooks/configuration/
```

The Ubuntu 16.04.1 LTS server has Python 3 by default, and hence we can use the same with Ansible. The *inventory/kvm/inventory* file contains the following:

```
[elovation-host]
ubuntu ansible_host=192.168.122.244 ansible_connection=ssh
ansible_user=ubuntu ansible_password=ubuntu123

[elovation-host:vars]
ansible_python_interpreter=/usr/bin/python3
```

You should now be able to issue commands, using Ansible, to the guest OS. For example:

```
$ ansible -i inventory/kvm/inventory ubuntu -m ping

ubuntu | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
```

Dependencies

The apt-daily.service runs by default on a new installation of Ubuntu 16.04.1 LTS. If we need to install software, the APT lock held by this service needs to be removed. Hence, we need to first stop this service. The APT software package repository is then updated before installing the dependencies to set up Ruby.

```
---
- name: Install dependencies
  hosts: ubuntu
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [apt]

tasks:
  - name: Stop apt-daily.service
    shell: systemctl kill --kill-who=all apt-daily.service

- name: Update the software package repository
  apt:
    update_cache: yes

- name: Install dependencies
  package:
    name: "{{ item }}"
    state: latest
  with_items:
```

```
- libssl-dev
- libreadline-dev
- zlib1g-dev
- build-essential
- git
```

The above playbook can be executed as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/elovation.yml --tags apt -vv -K
```

The -vv represents the verbosity of the Ansible output. You can use up to four 'v's. The -K option prompts for the sudo password for the *ubuntu* user.

Rbenv

Let's use Rbenv to set up Ruby on this virtual instance. It allows the installation of multiple Ruby versions, and the version in production can be selected. The source repo of Rbenv is cloned first, and then *configure* and *make* are executed. The Rbenv PATH and initialisation are then updated in the *~/.bashrc* file.

```
- name: Build rbenv
  hosts: ubuntu
  tags: [rbenv]

tasks:
  - name: Get rbenv
    git:
      repo: 'https://github.com/rbenv/rbenv.git'
      dest: "/home/{{ ansible_user }}/rbenv"

  - name: Build rbenv
    shell: "cd /home/{{ ansible_user }}/rbenv && src/
configure && make -C src"

  - name: Set rbenv PATH
    lineinfile:
      path: "/home/{{ ansible_user }}/bashrc"
      state: present
      line: "export PATH=\"$HOME/.rbenv/bin:$PATH"

  - name: rbenv init
    lineinfile:
      path: "/home/{{ ansible_user }}/bashrc"
      state: present
      line: "eval \"$(rbenv init -)\""
```

The playbook to set up Rbenv can be invoked as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/elovation.yml --tags rbenv -K
```